

Python Programming Learning Material

This PDF introduces the basic concepts of Python programming and can be used as learning material for generating an AI quiz.

1. Introduction to Python

Python is a high-level, interpreted programming language known for its simple and readable syntax. It is widely used in web development, data science, artificial intelligence, automation, and software development.

2. Variables and Data Types

Variables store data values. Common Python data types include int for integers, float for decimal numbers, str for text, bool for True or False values, list for ordered collections, tuple for immutable ordered collections, set for unique values, and dict for key-value pairs.

Example

```
x = 10
name = 'Python'
price = 99.5
is_active = True
```

3. Conditional Statements

Python uses if, elif, and else statements to make decisions. Conditions evaluate to True or False. Indentation is used to define blocks of code.

Example

```
age = 18
if age >= 18:
    print('Eligible')
else:
    print('Not eligible')
```

4. Loops

A for loop is commonly used to iterate over a sequence or range. A while loop repeats as long as its condition remains True. The break statement exits a loop, while continue skips the current iteration.

Example

```
for i in range(5):
    print(i)
```

5. Functions

Functions are reusable blocks of code defined using the def keyword. Functions can accept parameters and return values using the return statement.

Example

```
def add(a, b):  
    return a + b
```

6. Lists and Dictionaries

A list is an ordered and mutable collection. Items can be added using `append()`. A dictionary stores data as key-value pairs and values can be accessed using their keys.

Example

```
numbers = [1, 2, 3]  
numbers.append(4)  
student = {'name': 'Alex', 'age': 20}
```

7. Object-Oriented Programming

Python supports object-oriented programming through classes and objects. A class is a blueprint for creating objects. The `__init__` method is commonly used to initialize object attributes.

Example

```
class Student:  
    def __init__(self, name):  
        self.name = name
```

8. Exception Handling

Python uses `try` and `except` blocks to handle runtime errors. The `finally` block can be used for code that should execute whether an exception occurs or not.

Example

```
try:  
    number = int(input('Enter a number: '))  
except ValueError:  
    print('Invalid input')
```

Summary

Python provides readable syntax, dynamic typing, collections, control statements, functions, object-oriented programming, and exception handling. These concepts form the foundation for further Python development.